

ZACHARIAH DELLIMORE

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EDUCATION

Virginia Commonwealth University

Richmond, VA

Bachelor of Science in Computer Science, Magna Cum Laude, GPA: 3.89

May 2025

- Dean's List, 6 semesters.

PROFESSIONAL EXPERIENCE

Telus Digital

Sep 2025 – Present

Software Engineer

Charlottesville, VA

- Joined a project in crisis (20s+ average response times causing widespread user timeouts) and took ownership of performance work outside the assigned backlog.
- Reduced two high-traffic endpoints — a list query from 2.5s to 600ms (4x) and a single-record lookup from 2s to 50ms (40x) — and trimmed 10–25% off many other endpoints via targeted payload and query reductions.
- Proposed and prototyped a database schema redesign that tested at 5–10x improvement on high-cost endpoints (held for sequencing behind an in-flight migration).
- Co-presented these performance findings to the broader engineering org, sharing the optimization patterns we developed.
- Designed and built a Spring Boot service on GKE that monitors Telus's eBill notification pipeline end-to-end — Pub/Sub, BigQuery, GCS, Datastore, and Google Chat alerting on missed file arrivals, SLA breaches, and delivery failure spikes.
- Replaced 30+ minute daily manual checks and cut incident recovery from hours to minutes for a legally-mandated bill delivery process.
- Supported the Talent Program through career fair interviews and intern onboarding.

WillowTree Apps

May – Aug 2024

Return Software Engineer Intern

Charlottesville, VA

- Built 3 .NET REST API endpoints expanding reporting data available to client users, and enhanced existing endpoints to improve availability of critical reporting data.
- Used Azure Service Bus to efficiently transfer and replicate data between micro-services and their databases.
- Added authorization guards to existing endpoints to close a security gap.
- Led two client-facing demos showcasing new backend features.

PROJECTS

Game Boy Emulator

January 2026

- Built a Game Boy emulator in C++ and SFML that plays commercial ROMs.
- Implemented the full CPU instruction set, graphics rendering, memory mapping, and input handling, with savestates (custom serialization), adjustable game speed, and a configurable UI.
- Validated correctness against standard Game Boy hardware test ROMs (cpu_instrs, dmg-acid2) and a GTest unit suite organized by instruction category.

AI Web Game Generator

November 2025

- Won 2nd overall and 1st in the Gemini image generation (“nano banana”) category at the Cerebral Valley AIE Code Agents Hackathon — 2-person team.
- Built an AI agent that pulls game ideas off a queue and generates playable web games end-to-end: Claude generates the Phaser.js code while Gemini creates the in-game image assets.
- FastAPI backend orchestrating the agent loop, React/Vite frontend serving generated games, all packaged with Docker Compose.

Karaoke App

July 2024

- Built an automated karaoke app in C++/SFML that takes any audio file and uses Mozilla's DeepSpeech model to generate and display lyrics synced to the music in real time.

TECHNICAL SKILLS

Languages: C++, C, Go, C#, Java, Python, JavaScript, TypeScript, Kotlin, SQL, MATLAB, GLSL

Frameworks: .NET, Spring Boot, FastAPI, Node.js, Gin, React, Next.js, TailwindCSS, Jetpack Compose

Cloud & Tools: Google Cloud, Azure, AWS, Docker, Terraform, Git, CMake, JUnit, GTest, Claude Code, Codex